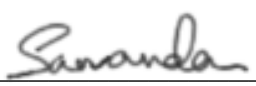


Project Report Design Project 1

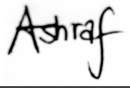
Project Title : Smart Hire

Team Members

1. Niaz Rahman 210042159
2. Antara Arifa Mullick 210042132
3. Labibah Hoque 210042121



Signature of Industry Supervisor
Fatema-Tuz-Zohora Sananda



Signature of Academic Supervisor
Ashraful Alam Khan

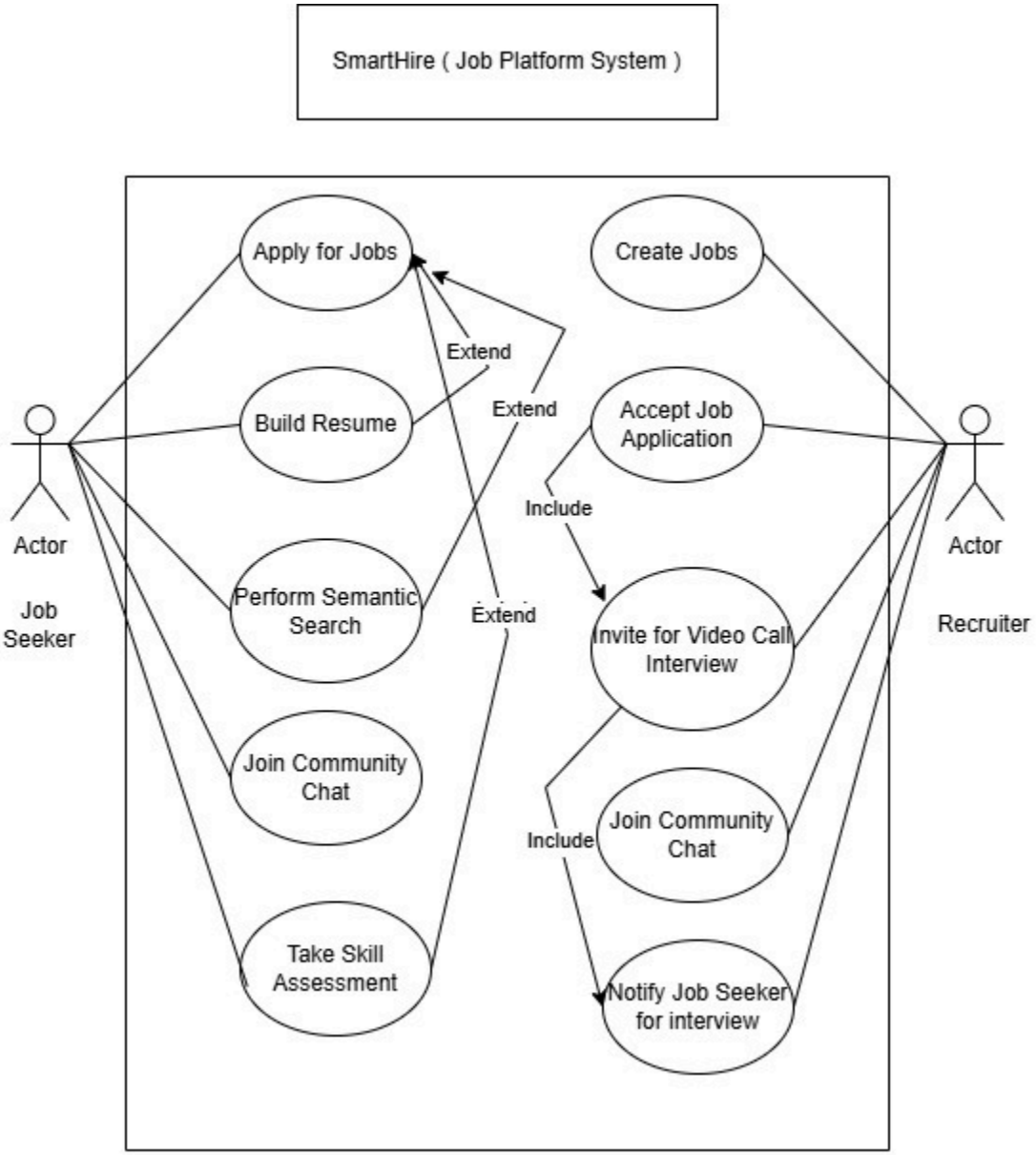
Project Overview

SmartHire is an innovative digital platform developed to address the growing challenges in Bangladesh's job market by connecting job seekers with employers efficiently. Designed as an all-in-one solution, the platform offers a range of features tailored to both job seekers and recruiters. For job seekers, SmartHire provides tools such as an AI-powered resume builder, skill assessment quizzes, semantic job search, and a community chat for networking. Recruiters benefit from streamlined job posting, applicant management, and integrated video interviews, making the hiring process faster and more effective. By leveraging modern technologies like React.js, Node.js, and AI-driven semantic search, SmartHire aims to reduce unemployment by bridging the gap between workforce skills and employer demands. The platform not only enhances job search efficiency but also fosters a supportive ecosystem where candidates can improve their employability and recruiters can find the right talent with ease.

Motivation Behind the Project

The motivation behind SmartHire stems from the pressing issues in Bangladesh’s labor market, where unemployment is rising (currently at 4.20%) and skill mismatches leave nearly half of employers struggling to find qualified candidates. With over 4.56 million unemployed individuals, many face rejection due to poorly crafted resumes or inadequate interview preparation, issues that SmartHire directly addresses. Additionally, traditional hiring methods are often slow and inefficient, creating barriers for both job seekers and employers. Recognizing these challenges, SmartHire was developed to modernize the recruitment process. By integrating AI-driven tools, real-time communication features, and skill assessments, the platform empowers job seekers to present themselves effectively while helping recruiters identify the best candidates quickly. Ultimately, SmartHire seeks to contribute to economic growth by making job markets more accessible, efficient, and aligned with the needs of today’s workforce.

Use Case Diagram



Features of the Project

Job Seeker Features

1. **Apply for Jobs**
   - ◯ Users can browse and apply for job listings posted by recruiters.
2. **Resume Builder**
   - ◯ Helps users create professional resumes by filling in their details, skills, education, and work experience and leveraging AI to enhance the CV
3. **Community Chat**
   - ◯ A chat feature where job seekers can join groups, discuss job-related topics, and seek advice from others.
4. **Semantic Search**
   - ◯ An advanced search tool that helps users find relevant job postings based on keywords and meanings, not just exact matches.
5. **Skill Assessment**
   - ◯ **Users can take quizzes to test their skills in areas like Frontend Development, Backend Development, or Data Science.**

Recruiter Features

6. **Post Jobs**
   - ◯ Recruiters can create and publish job listings with details like title, description, salary, and requirements.
7. **Accept Job Applications**
   - ◯ Recruiters can review and accept or reject applications submitted by job seekers.
8. **Invite for Video Interview**
   - ◯ Recruiters can schedule and invite candidates for video interviews directly through the platform.
9. **Community Chat**
   - ◯ Recruiters can also participate in group chats to interact with job seekers or other recruiters.

General Features

10. **Notification System**
    - ◯ Job seekers get alerts when their application is accepted or rejected

Tools and Technology

Frontend Development

- **React.js:** A JavaScript library for building dynamic and responsive user interfaces.
- **Tailwind CSS:** A utility-first CSS framework for rapid UI development.

Backend Development

1. **Node.js:** A runtime environment for server-side JavaScript execution.
2. **Express.js:** A web application framework for building RESTful APIs.

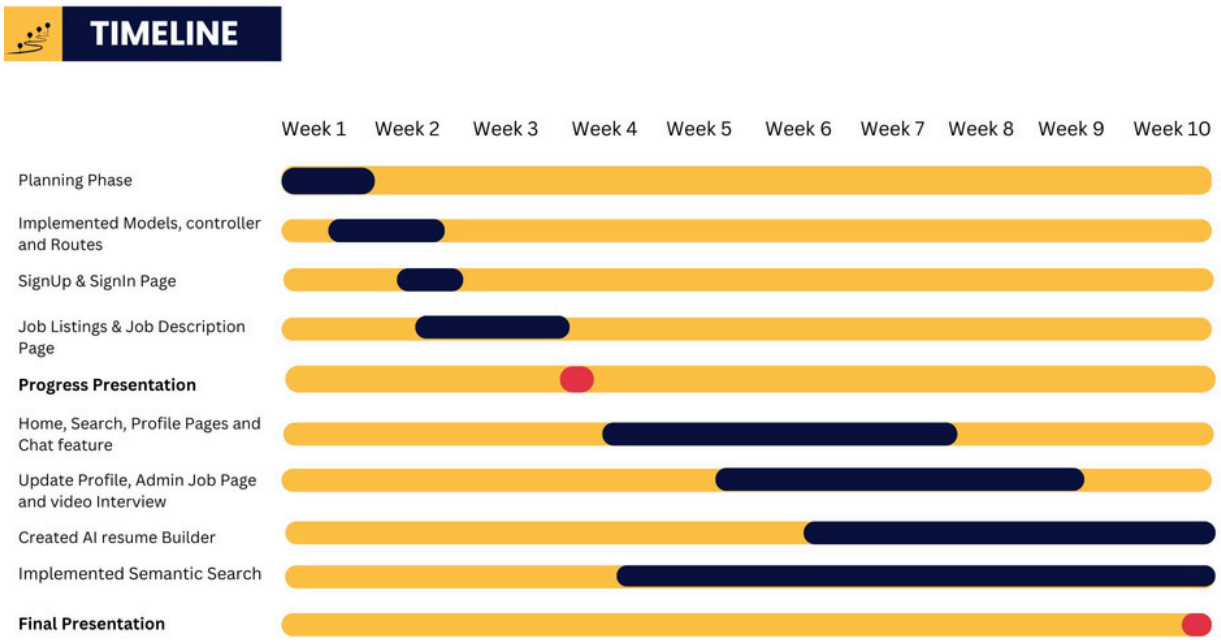
Database & Version Control

- **MongoDB:** A NoSQL database for flexible and scalable data storage.
- **GitHub:** For collaborative version control and project management.

AI & API Integration

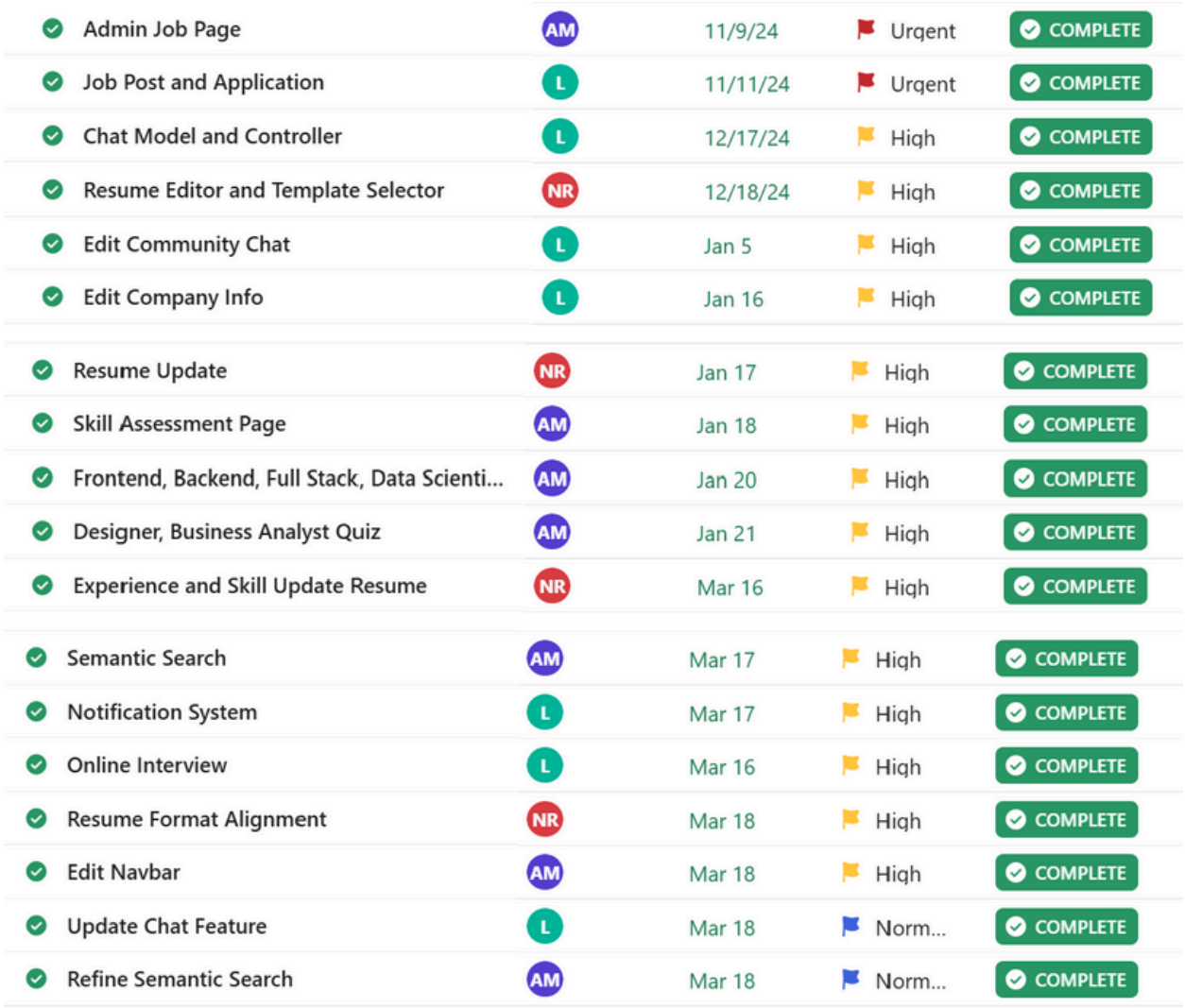
3. **Groq: Used for AI-powered resume-building assistance.**

Timeline



Product Backlog:

Name	Assignee	Due date	Priority	Status
User, Company, Job Model	AM	10/4/24	Urgent	COMPLETE
User, Company, Job Controllers and Routes	AM	10/12/24	Urgent	COMPLETE
Application Model, Controllers and Routes	AM	10/21/24	Urgent	COMPLETE
SignUp and SignIn page	NR	10/30/24	Urgent	COMPLETE
Home page	NR	11/5/24	Urgent	COMPLETE
All Job Page,Job Description Page	L	11/4/24	Urgent	COMPLETE
Search Page, Profile page	L	11/5/24	Urgent	COMPLETE
Update Profile	NR	11/6/24	High	COMPLETE



Future Scopes:

1.Job Recommendation:

- Recommend Jobs for job seekers based on their skill assessment, search history in the website and interests.

2.Update Profile:

- Add the skill assessment score in the profile of job seekers.

Project Repository Link : https://github.com/AntaraArifa/SmartHire.git